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Studierichting: Informatica
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$$\int \frac{2x^2 - x + 1}{2x + 1} dx$$

$$t = 2x + 1 \Rightarrow dt = 2dx \Rightarrow \frac{1}{2}dt = dx$$

$$x = \frac{t-1}{2}$$

$$= \int \frac{2\left(\frac{t-1}{2}\right)^2 - \frac{t-1}{2} + 1}{t} \cdot \frac{1}{2}dt$$

$$= \int \frac{\frac{(t-1)^2}{2} - \frac{t-1}{2} + 1}{t} \cdot \frac{1}{2}dt$$

$$= \int \frac{\frac{(t-1)^2 - (t-1) + 2}{2}}{t} \cdot \frac{1}{2}dt$$

$$= \int \frac{\frac{t^2 - 3t + 4}{2}}{t} \cdot \frac{1}{2}dt$$

$$= \int \frac{(t^2 - 3t + 4) \cdot \frac{1}{2}}{t} \cdot \frac{1}{2}dt$$

$$= \frac{1}{4} \int \frac{t^2 - 3t + 4}{t} dt$$

$$= \frac{1}{4} \int t - 3 + \frac{4}{t} dt$$

$$= \frac{1}{4} \left(\int t dt - 3 \int 1 dt + 4 \int \frac{1}{t} dt \right)$$

$$= \frac{1}{4} \left(\frac{1}{2}t^2 - 3t + 4 \ln |t| \right) + C$$

$$= \frac{1}{8}t^2 - \frac{3}{4}t + \ln |t| + C$$

$$= \frac{1}{8}(2x + 1)^2 - \frac{3}{4}(2x + 1) + \ln |2x + 1| + C$$

References